SDG BLOCKCHAIN ACCELERATOR

Market Penetration and Scaling Strategy

Cubid Protocol - Solar Village

(Nigeria)

This document outlines the market penetration strategy, branding approach, user acquisition plan, go-to-market strategy, and long-term scaling roadmap for the Cubid protocol pilot developed within the SDG Blockchain Accelerator program.

1. Project Overview

The Solar Village pilot, powered by the CUBID Protocol, addresses a critical energy challenge in rural Nigeria. Millions of people in remote communities lack reliable access to electricity and rely on expensive, polluting diesel generators and kerosene for power. Solar mini-grids have been deployed as a cleaner alternative, but their long-term viability has been limited by weak investment models, ad-hoc payment collection, poor financial records, and a lack of transparency. Mini-grid operators struggle to demonstrate predictable revenues, governments and development partners lack reliable data for oversight, and investors face high verification costs. As a result, even when solar infrastructure is installed, it often fails to scale, carbon finance remains out of reach, and rural households continue to face high energy costs and unreliable service. The core problem is not generating electricity per se, but the absence of a trusted, transparent, and inclusive operating and reporting infrastructure to support these rural mini-grids.

The Solar Village solution directly tackles this gap by introducing a blockchain-based operations platform for rural solar mini-grids. In essence, Solar Village is a blockchain-enabled management system that creates tamper-evident energy and payment records for off-grid solar installations. The platform integrates with the mini-grid infrastructure to automatically log electricity generation and consumption data at each site, recording it in a secure per-mini-grid ledger using Cardano's Hydra protocol. Each day, aggregated proofs of the energy data are anchored to the Cardano mainnet, producing an immutable audit trail of every kilowatt-hour produced and consumed. This ensures that electricity delivery and payments are transparent to all stakeholders, from local operators and community cooperatives to national regulators and development partners, and lays the groundwork for credible carbon credit verification based on real, verifiable energy data. In practical terms, the system links digital payment channels (including mobile-based wallets and USSD for basic phones) with smart meter data, so that every payment for energy is recorded as a tokenized transaction on the ledger. Energy credits and payments are represented as on-chain tokens, secured by the blockchain, which creates a permanent, tamper-proof record of consumption and revenue flows. This token flow – from a resident's digital wallet top-up to the mini-grid operator's account – brings unprecedented visibility into household payments and the utilization of funds, while also enabling future integration with carbon markets (e.g. tokenized carbon credits once emissions reductions are verified).

The pilot is a collaborative effort led by UNDP Nigeria in partnership with CUBID Protocol and other stakeholders. The concept for Solar Village was incubated within UNDP Nigeria's digital innovation program, and the initiative continues to be closely supported by the UNDP country office. CUBID Protocol, as the solution maker, contributes its expertise in decentralized identity management and tamper-evident transaction technology to ensure secure user authentication and data integrity within the platform. Together with national agencies and experts, the team has developed the Solar Village platform as a piece of digital public infrastructure rather than a private product. Notably, the pilot project builds on existing rural electrification initiatives instead of creating parallel infrastructure. It is designed to augment the operations of solar mini-grids already deployed or planned in Nigeria. By focusing on operational transparency, efficient tariff collection, and performance tracking, Solar Village complements the physical electrification work of programs under Nigeria's Rural Electrification Agency (REA) and others, rather than duplicating those efforts. Key beneficiaries of the pilot include local households and community cooperatives (who gain more reliable service and potentially lower costs), solar mini-grid operators (who gain a robust system for revenue collection and reporting), and oversight bodies like REA and the National Council on Climate Change (which gain access to trustworthy data for policy and carbon finance). In summary, this pilot demonstrates how emerging technologies, specifically blockchain and digital identity, can be applied in a community-driven energy context to improve transparency, accountability, and sustainability of rural electrification projects in Nigeria.

2. Market Penetration Strategy

The market penetration strategy for Solar Village begins with a focused deployment in a defined segment of Nigeria's off-grid energy market and then scales out in stages. In the immediate term, the project leverages a "captive" base of 23 rural mini-grids in Nigeria as pilot sites, secured through partnership with the Rural Electrification Agency (REA). These 23 pilot mini-grids serve roughly a few hundred households in total (approximately 200+ households across the villages) and provide an ideal initial user base for the platform. Importantly, these communities were selected in collaboration with REA, which oversees rural electrification in Nigeria, ensuring that the pilot is embedded in national programs from the start. Many of these existing mini-grids currently lack any integrated digital payment or monitoring solution, tariff collections are often handled in cash or through ad-hoc methods, and operational data is reported manually. Solar Village addresses these pain points by introducing built-in mobile payment options (including support for local payment agents who can facilitate transactions on behalf of villagers) and by automating the required reporting to authorities. By immediately solving real operational gaps, such as simplifying how villagers pay for power and how operators report performance, the platform is positioned to achieve high adoption and engagement at these pilot sites.

Once the pilot phase proves the concept and demonstrates value, the strategy shifts to scaling up within Nigeria. The next step is to convert this initial success into broader market penetration by targeting the wider pool of rural mini-grids and off-grid communities that face similar challenges. Nigeria's market for rural energy is vast: an estimated $14 billion is spent annually on diesel fuel for generators in the country, representing a huge opportunity for solar alternatives. Every kilowatt-hour delivered by solar in place of diesel not only reduces carbon emissions but also taps into this "wasted" diesel expenditure as potential revenue for sustainable energy services. By offering a proven operations platform, Solar Village can help channel even a fraction of this spending into clean energy usage – a shift that would translate into substantial growth. In practical terms, the team plans to scale by enrolling additional mini-grid operators into the platform. Through our partnerships, we have insight into a pipeline of new and existing mini-grids across Nigeria; for instance, REA is overseeing hundreds of deployments nationwide and has identified an initial 23 for our pilot, with many more to follow as the solution is validated. We are also collaborating with the African Minigrid Program (AMP), a regional initiative, to integrate Solar Village into ongoing donor-supported electrification projects. This partnership-based approach acts as a force multiplier: REA and AMP serve as institutional anchors that directly connect us to rural electrification efforts and communities across Nigeria. As the pilot results materialize, we will work with these agencies to expand the platform to additional villages in their portfolios (for example, onboarding new mini-grids that REA or AMP bring online in the next year). The medium-term goal (6–18 months) is to replicate the Solar Village model in a significant share of Nigeria's mini-grids, scaling up the user base from the initial few hundred households to several thousand across different states. During this phase, we aim to demonstrate how the platform improves collection efficiency, increases transparency, and potentially enables access to carbon credits, thereby creating a strong value proposition for operators and community stakeholders in each new site.

Looking beyond Nigeria, Solar Village is designed to transition from a pilot of 23 sites to a scalable Software-as-a-Service (SaaS) model accessible to thousands of mini-grid operators across Africa. The long-term vision aligns with the broader "Mission 300" initiative targeting 300 million Africans for energy access – effectively positioning Solar Village as a digital backbone for off-grid energy markets at continental scale. In practical terms, this means that after firming our foothold in Nigeria, we plan to extend the platform to other countries with similar rural electrification challenges. Our open technology architecture supports this expansion: Solar Village is intended to become an open-source SaaS platform that can be adopted and localized by different partners and communities across Africa. Market penetration in new regions will again leverage partnerships – for example, engaging with national rural electrification agencies, international donors, and private mini-grid developers in target countries to integrate the solution into their projects. By highlighting the success metrics from Nigeria (such as improved payment recovery rates, reduced operational overhead, and enabled carbon revenue), we will

build trust with new stakeholders. The scalability of the platform has been a priority from the start: it is cloud-based and modular, capable of onboarding new sites rapidly without extensive on-ground infrastructure beyond connecting the meters. Moreover, the value proposition is compelling and universal: any mini-grid operator or investor, in Nigeria or elsewhere, benefits from better transparency and potential new revenue streams (like carbon credits) that Solar Village provides. In summary, our market penetration strategy begins with a deep focus in Nigeria's rural communities – proving the model in partnership with government programs – and then expands outward by converting that proof into a pan-African offering, ensuring that hundreds of off-grid communities can be reached through a combination of direct partnerships and a scalable SaaS deployment model.

3. Branding and Positioning

3.1 Brand Identity

The project's branding centers on the concept of "Solar Village" – a name that immediately conveys community-scale solar empowerment. This branding was deliberately chosen to be inclusive and aspirational: it speaks to the transformation of villages through solar technology, emphasizing local development and sustainability. All communications present Solar Village not just as a tech platform, but as a community-centric solution that bridges clean energy and innovation. The imagery and narrative focus on rural households gaining reliable power, children doing homework under electric light, and community members engaging with technology that improves their lives. This human-centered storytelling is key to positioning the solution in a positive light for both the community beneficiaries and the supporting stakeholders. We have developed accessible communication materials, including a one-page graphical summary and a pitch deck, to promote the solution's vision and impact in simple terms. These materials highlight the core benefits – affordable clean energy, transparency, and new income opportunities (via carbon credits) – in a way that resonates with non-technical audiences and decision-makers alike. By using clear visuals and local context (such as photos of the pilot community and testimonials), the branding reinforces that Solar Village is *for* the people it serves and is co-created with their needs in mind.

3.2 Positioning in the Ecosystem

Strategically, Solar Village is positioned as an *enabling infrastructure* rather than a competitor to existing energy providers. We emphasize that our solution **complements existing rural electrification initiatives by strengthening operational viability, not by duplicating or replacing physical infrastructure**. This positioning has been communicated to government partners and mini-grid operators to ensure we are seen as collaborators. For instance, in conversations with

REA and local operators, Solar Village is presented as a toolkit that can enhance their projects' success – by improving revenue collection, providing data for reporting and grants, and opening access to carbon finance – rather than an external project with a separate agenda. This collaborative stance is further bolstered by UNDP's involvement. Being an initiative that originated within UNDP Nigeria gives Solar Village a high degree of credibility and neutrality. We have effectively branded it as a UNDP-supported innovation in sustainable energy, which assures communities and partners that the project's motive is development impact, not profit. At the same time, the cutting-edge technology aspect (blockchain and digital identity via CUBID Protocol) positions Solar Village at the forefront of innovation. We highlight that **CUBID Protocol's expertise in decentralized identity and secure transactions underpins the platform**, signaling that while the solution is community-focused, it is built on robust, state-of-the-art technology. This dual identity – as both a grassroots development solution and a high-tech innovation – is a unique selling point. It lets us engage a wide range of stakeholders: from rural community leaders (who see the tangible benefits for their village) to technologists and investors (who see a scalable, innovative product with global relevance).

Our positioning also stresses **transparency and openness**. We articulate that Solar Village is essentially digital public infrastructure for energy. In line with that, we have plans to open-source the platform after the pilot stage, inviting wider collaboration and trust. By aiming to become an open-source SaaS platform, we differentiate Solar Village from proprietary energy management software – our goal is to be a common good that any mini-grid operator or government can use and adapt across Africa. This open approach is part of the brand ethos: transparency in the code and data mirrors the transparency in energy usage that we deliver. It positions the project as aligned with global open innovation and "digital public good" principles, which can attract support from institutions that prefer solutions without vendor lock-in.

Finally, the branding is strongly linked with partnerships for legitimacy. We consistently communicate that Solar Village is backed by a coalition of reputable partners: the Nigerian government (REA and the Climate Change Council), international organizations (UNDP), technical experts (like the Solar Foundation and M3tering Protocol), and the Cardano blockchain community. These partners **provide sector legitimacy and policy alignment**, effectively endorsing Solar Village's mission. For example, featuring REA's and UNDP's logos in our materials and mentioning their roles gives confidence to other mini-grid developers or community leaders that this is an officially supported, serious initiative. This multi-stakeholder backing not only strengthens the brand's trustworthiness, it also signals that Solar Village is woven into larger frameworks (like Nigeria's energy transition plan and carbon market development). In summary, our branding and positioning strategy presents Solar Village as a **trustworthy, inclusive, and forward-thinking platform** – one that is community-first in its

narrative, collaborative in its ecosystem role, and innovative and open in its technological identity.

4. User Acquisition and Go-To-Market Strategy

Our user acquisition strategy is two-fold, reflecting the dual user base of the Solar Village platform: (1) the **mini-grid operators and program implementers** who are our direct clients, and (2) the **community end-users** (households and local businesses) who ultimately use and benefit from the electricity and the payment system. The go-to-market approach carefully addresses both groups to drive adoption.

For mini-grid operators and implementation partners, the strategy is to demonstrate clear value and make onboarding seamless. Early in the pilot, we engaged a few operators as co-design partners – they provided input during development and got to see the platform's benefits firsthand. By incorporating their feedback, we ensured the system addresses real operational needs (for example, providing dashboards that simplify their monitoring and reporting tasks). The result is that operators see Solar Village as solving pain points rather than adding work. **During testing, the dual focus on operator financial viability and household accessibility was highlighted as a major strength of the solution** – operators appreciated that the platform helps them secure revenue and data, while also being user-friendly for their customers. This gives us a strong word-of-mouth advantage: operators involved in the pilot can advocate for the platform to peers. As we move to scale, we will leverage industry forums and networks (such as mini-grid developer associations, REA's stakeholder workshops, and AMP meetings) to showcase the pilot results. A key part of our go-to-market pitch to operators is the **increased transparency and potential new revenue streams** (like carbon credits) that Solar Village unlocks. We will present concrete evidence from the pilot, such as improved payment collection rates and the creation of verifiable energy records ready for carbon finance, to convince other operators that adopting our system has direct financial and compliance benefits. To lower barriers to adoption, our model is to offer the platform with **minimal upfront cost during the initial rollout phase**, especially if the operator is part of a government or donor program. This can be done thanks to grant funding covering initial deployment. Only once the value is proven do we introduce a modest subscription or fee (as detailed in the Financial Sustainability Model). This approach, essentially a free trial/pilot period, is crucial to acquiring users in a conservative sector – it de-risks the decision for operators. Additionally, integration with existing processes is emphasized: because REA requires monthly reports from operators, and our platform automates that reporting, we position Solar Village almost as a *compliance facilitator*. An operator who uses Solar Village can essentially fulfill regulatory reporting obligations at the click of a button, which is a strong incentive for them to come onboard. We intend to formalize such advantages in partnerships: for instance, we will seek REA's

endorsement so that using Solar Village could become a recommended (or even standard) approach for mini-grid reporting. This kind of institutional support would significantly accelerate user acquisition among operators.

For community end-users, our strategy focuses on **accessibility, trust, and local engagement**. Most villagers are not familiar with blockchain or even mobile banking, so the system must meet them where they are. To that end, we have implemented a very user-friendly payment experience: consumers can purchase electricity credits through **familiar channels like USSD codes on basic phones or through local agents in their village**. This means a customer can walk to a nearby shop (or agent) and pay in cash or use a simple phone menu to top up their meter, much like buying mobile airtime – a process they already know. The platform was explicitly designed to enable prepaid electricity via these familiar methods in order to **promote financial inclusion** and ease of use. Each user is provisioned a lightweight digital wallet (though they might not even realize it's a "wallet" in blockchain terms) that holds their energy credits. The experience is kept intuitive: once payment is made, either the meter is remotely activated or a code is provided to the user to enter into their meter to unlock the paid amount of kWh. Through the pilot, we are actively engaging with the community to build trust in this system. We conduct in-person demonstrations and training sessions in the village, often in collaboration with the mini-grid operator's staff who are known to the community. We also rely on **community cooperatives and local leaders** to spread awareness. For example, during the rollout, our team organized community meetings where we explained the new payment process, how it ensures everyone's payments are recorded and acknowledged, and how it could even help attract improvements (like more solar capacity or stable tariffs) through its transparency. This grassroots engagement is an ongoing process – we have built in feedback loops to listen to user concerns and adapt. During the pilot's initial weeks, **short interviews and community check-ins were held to gather user reactions**, and even simple surveys were administered via USSD to gauge satisfaction. Operator agents on the ground noted any difficulties residents had, such as if anyone struggled with the USSD steps or had questions about their balance. This feedback was invaluable; in fact, it led us to make several user-centric improvements. One example is **simplifying the sign-up and onboarding flow for new users** – based on early feedback, we reduced the number of steps and made the interface (and USSD prompts) more straightforward so that even those unfamiliar with mobile apps could register or be registered easily. Another example is expanding payment options: originally we focused on mobile money and agent payments, but users indicated a preference for using their ATM/debit cards as well, so we updated the system to accommodate debit card payments for buying energy credits. By reacting quickly to community input, we show the users that this solution is *for* them and is being shaped *with* them, which greatly enhances acceptance.

Our go-to-market strategy in the communities also involves identifying and empowering **local champions** – typically, a member of the community (perhaps a tech-savvy youth or a respected teacher) who can act as a liaison or support for others. We train these individuals as "Solar Village agents" who understand the platform and can assist their neighbors in using it, especially in the early stages. This not only creates local jobs (a form of community benefit) but also ensures sustainability of user support without our team's constant presence. Additionally, we plan to coordinate with local government and traditional authorities to get their buy-in. Having the endorsement of a village chief or the local government council can significantly boost community trust in the new system. We have been careful to frame the project as a community development initiative (which it is), sometimes even co-branding with government rural electrification initiatives, so that users perceive it as an officially backed improvement to their service, not an outside experiment.

From a broader perspective, the **go-to-market timeline** aligns with the phased expansion described earlier. In the immediate pilot phase, our acquisition is very hands-on and localized. As we move to the **post-accelerator scale-up**, user acquisition will rely more on network effects and partnerships. For instance, once we have success stories from the pilot villages (e.g. increased revenue collection by X% or village electrification rates improving), we will publish these case studies and work with REA's communications team to highlight them nationally. This can attract inbound interest from other communities or mini-grid operators who hear about the benefits. We are also exploring integration into broader platforms – for example, if Nigeria launches any digital marketplace or registry for carbon credits or for mini-grid performance, we want Solar Village to be compatible or listed as a service. Such integration means any operator joining those programs could easily onboard with us as well. In terms of channels, beyond direct partnership outreach, we will utilize the Cardano and blockchain community networks (since we are a Cardano-based project, the Catalyst community and Cardano social channels can amplify our message to other innovators or funders who might help us reach new markets). We have already applied to the Cardano Accelerator Program to gain further exposure. In summary, the user acquisition and go-to-market strategy begins with **high-touch engagement in pilot communities and close collaboration with initial operators**, ensuring a successful proof-of-concept. Then it evolves into a more scalable model where **institutional partnerships, success narratives, and an open, user-friendly platform** drive the adoption across many communities with less direct intervention needed per site.

5. Post-Accelerator Sustainability and Scaling Roadmap

With Phase 2 of the accelerator completed and the pilot-ready solution in hand, our focus turns to sustaining the initiative and scaling it up in a phased manner. The **Post-Accelerator Roadmap**

is organized into short-term, medium-term, and long-term actions, each building on the success of the previous stage.

5.1 Short Term (0–6 months)

Pilot Execution and Validation: In the immediate months after the accelerator, the priority is to execute the pilot project on the ground and validate the solution in real conditions. This involves finalizing any remaining system integrations (such as connecting to actual smart meters and payment gateways in the field) and deploying the Cardano Hydra heads and necessary backend infrastructure for each pilot mini-grid. By deploying the full technology stack in the pilot communities, we will generate live data and demonstrate the platform's functionality under real load. Alongside the technical deployment, we will launch the operational dashboards for the mini-grid operators and oversight bodies (e.g. UNDP, REA) so that they can start monitoring performance in real-time. The pilot will then go live with our confirmed partners – meaning the 23 selected mini-grids will begin using Solar Village for their day-to-day energy sales and data logging, under the supervision of REA, AMP, and the project team. During this phase, sustainability is about making sure the solution truly works and delivers value: success will be measured by indicators like system uptime, user adoption rates in the community, operator satisfaction, and data reliability. We have allocated resources for on-site support and troubleshooting throughout this period. The outcome we seek is a validated model – essentially proof that if Solar Village is implemented in a rural community, it achieves the intended improvements (efficient payments, transparency, etc.) without major issues. Achieving this will set the stage for scaling.

5.2 Medium Term (6–18 months)

Scale Readiness and Expansion: Once the pilot is proven, the next 1 to 1.5 years will focus on preparing the solution for broader deployment and actually scaling it to more sites. A primary goal in this phase is to **expand to additional mini-grids within our existing partner networks in Nigeria**. Practically, this means working with REA and AMP to roll out Solar Village to other communities that they support. For example, if REA's program is commissioning 50 new mini-grids in the next year, we aim to integrate our platform into as many of those as possible. We will also reach out to private mini-grid developers who were not part of the initial pilot but operate in similar regions, offering them an opportunity to onboard. To handle this growth, we will improve and standardize our onboarding process (documentation, training materials, possibly creating a "playbook" for new deployments). On the technical side, the medium term will see the platform hardening and feature refinement needed for scale: we will strengthen reporting and audit tools to handle a larger user base and more data. A crucial aspect of sustainability here is financial and institutional integration. In the medium term, we plan to **introduce structured pricing for operators** – in other words, transitioning from free pilot usage

to a sustainable business model where operators pay a subscription or fee for the service. This pilot-to-commercial transition will be done gradually; we might start with a nominal fee or a tiered model as incentives, ensuring it does not hinder adoption. The introduction of pricing will coincide with demonstrating clear ROI to the users (e.g. showing that their increase in revenue or decrease in overhead thanks to Solar Village far exceeds the fee). Alongside this, we will formalize our offering in institutional terms: possibly signing MOUs or agreements with government agencies or a hosting arrangement for the platform as part of a national energy data system. We will also **align with national and international carbon measurement standards** during this period. This means working with the National Council on Climate Change (NCCC) to ensure that the data our system collects can directly feed into Nigeria's carbon credit framework. By the end of 18 months, we expect to have not only a larger footprint in Nigeria (potentially dozens of communities using the platform), but also the frameworks in place for carbon credit issuance (e.g. methodologies agreed, data pipelines to NCCC established), and initial revenues flowing in from our service model. Our team will likely grow in this phase, adding dedicated support and deployment personnel to handle multiple sites concurrently. Importantly, we will be assessing which parts of the model can be templated for replication beyond Nigeria, even as we focus on the immediate expansion at home.

5.3 Long Term (18+ months)

Sustainable Scale and Regional Expansion: In the longer horizon, our roadmap envisions Solar Village becoming a self-sustaining, widely adopted platform across multiple regions. **One major objective is a national or multi-region rollout in collaboration with government programmes**. In Nigeria, this could mean institutionalizing Solar Village as part of the government's rural electrification toolkit – for example, making it the recommended (or required) system for any mini-grid receiving public funding or tariff subsidies. We would work with REA and possibly the Ministry of Power to achieve this level of integration. Concurrently, we aim to expand to other countries in sub-Saharan Africa. Using the open-source SaaS approach, we can partner with organizations like UNDP offices in other countries, regional development banks, or pan-African programs to introduce Solar Village to new markets. By 18+ months, we plan to have at least pilot engagements or initial deployments beyond Nigeria, in line with the Mission 300 vision. Another key milestone in the long term is to **launch full carbon credit issuance workflows** where applicable. If the medium term established the data alignment and verification processes, the long term will see the first carbon credits being issued based on Solar Village's data, potentially generating revenue for the communities and operators. This will be a significant proof point for the model's impact (turning verifiable clean energy into cash via carbon markets) and will attract further investment and participation. In terms of service delivery, by this stage we will refine our offering into a more formal product – potentially **offering the platform as a managed SaaS or as a public digital infrastructure service** available

openly. This means we will either continue to operate it as a company/consortium providing services to clients, or we will have transitioned it to an open platform that an ecosystem of vendors and users can operate (or some hybrid of both). Determining the exact modality will depend on what ensures the most sustainable growth – we remain open to possibly entrusting a governmental body or an alliance with running the core infrastructure if that makes sense for longevity, while we focus on innovation and support. By pursuing open-source, we ensure that no matter what, the platform can live on and be adapted even beyond the founding team's direct efforts, which is important for true sustainability.

Underpinning this roadmap is a **sustainability plan** that covers resources and partnerships. Financially, our projections indicate that to support a regional rollout (dozens of sites, thousands of users), we will need an operational budget on the order of $250k–$400k per year, and for pan-African scale (hundreds of sites) around $500k+ per year. These costs include maintaining cloud infrastructure, ongoing development, customer support, and compliance work. In the post-accelerator stage, we plan to meet these needs through a combination of revenue (from the platform's fees) and external funding. In the short to medium term, before the platform's own revenue is sufficient, we will rely on **blended finance** – for example, continuing to secure grants or concessional funding, and in-kind support from partners to subsidize operations. We have a fundraising plan targeting various sources at each stage: in the short term, innovation grants from UNDP or climate-focused funds; in the medium term, climate finance facilities and development banks as we scale; and in the longer term, impact investors and results-based financing tied to our outcomes. We will kickstart these efforts immediately after the pilot demonstrates results, using the evidence to reduce the perceived risk for funders. On the partnership front, we will continue strengthening ties with institutional partners like REA, NCCC, and AMP for policy support and integration. We also plan to formalize partnerships with technical organizations for metering and data (to keep improving the tech) and possibly with private sector players like telecom companies if needed for USSD services. By maintaining this broad support network, we ensure that Solar Village is not solely dependent on one entity but is embedded in a larger ecosystem committed to its success.

In summary, our post-accelerator roadmap is a clear, phased journey: **validate locally, expand nationally with structure, then replicate regionally with sustainability**. Each phase has defined goals – from execution and proving value, to scaling and monetizing, to broad adoption and integration into systems – all aimed at making Solar Village a lasting solution for rural energy challenges well beyond the accelerator program.

6. Financial Sustainability Model

Achieving financial sustainability is crucial for Solar Village to thrive beyond the initial pilot and grant-funded stage. The financial model is built on transitioning from purely donor-funded operations to a mix of **earned revenues and strategic funding** that covers costs as the platform scales. In the post-pilot phase, once we have demonstrated the platform's value, we will begin to generate income through several channels:

- **Subscription Fees from Operators:** Mini-grid operators and energy service companies using the platform will pay a subscription or license fee for access to Solar Village's services. This fee grants them continued use of the dashboards, reporting tools, and maintenance/support of the system. It is essentially a SaaS fee for the software. We intend to keep this affordable, tiered by the size of the operator or number of connections, to encourage uptake. As noted, features like reporting and compliance tools are provided, which justify the cost. According to our Phase 2 plan, these **subscription fees paid by mini-grid operators** will cover platform access, data reporting, and compliance-related tooling that we offer.

- **Transaction/Usage Fees:** In addition to (or in some cases instead of) fixed subscriptions, the platform can charge small **usage-based fees tied to the energy volume transacted or value-added services provided**. For example, there could be a fee of a few cents per kWh recorded or a percentage fee on each payment transaction processed through the platform. Another model is charging for premium services, such as advanced analytics or custom integrations. This usage-based approach aligns our revenue with the success of our users – as operators sell more power (and presumably earn more), our income would increase correspondingly. It also lowers the barrier for very small installations to join (since if usage is low, their fee is low). We will calibrate these fees carefully to ensure they do not discourage use; the aim is they are marginal relative to the benefits (for instance, if we help recover revenue that would otherwise be lost, a small transaction fee is acceptable). The Phase 2 report highlights this as a component of the model, indicating **transaction fees tied to metered energy volume or services** as a revenue source.

- **Carbon Credit Revenue Share:** A unique and potentially significant future income stream comes from carbon finance. Once the platform is facilitating verified carbon credit issuance (for the diesel fuel that solar replaces), **Solar Village could earn a portion of the carbon credit revenue or fees for enabling Monitoring, Reporting, and Verification (MRV)**. In practice, this might work as follows: for every ton of CO_2 emission reduction that is certified via data from our platform, a carbon credit is sold (to, say, a corporate buyer or on a carbon market). The revenue from that sale largely goes to the mini-grid operator or project owner (benefiting the community by extension), but the

platform takes a small percentage or flat fee for having provided the trusted data backbone that made the credit possible. This aligns incentives – we want to help operators earn carbon income, and we only earn more if they do. We have already aligned with the NCCC on future carbon credit integration, so when this capability comes online, it can become a win-win source of funds.

- **Institutional Licensing and Service Agreements:** We will also pursue enterprise-level agreements, for example with government agencies, development programmes, or NGOs that might want to deploy the solution at scale. In such cases, rather than charging per site, we might have an **institutional licensing model** where a government pays for an entire region's mini-grids to be connected, or a donor program pays us to implement and support Solar Village for a cohort of communities. Under this model, our platform can be white-labeled or co-branded if needed, and we provide training and support as a service. This could even take the form of a *public-private partnership* where the government contracts us to run the digital infrastructure of rural electrification (making it effectively a long-term service contract). The Phase 2 plan explicitly includes **service agreements with governments or development programmes** as part of the revenue model mix.

By diversifying these income streams, we ensure that Solar Village isn't reliant on any single source of revenue. Some streams (like subscriptions) will start soon after pilot, while others (like carbon credit revenue) will kick in later once the enabling mechanisms are in place. According to our staged approach, **revenue generation is expected to begin only after pilot validation**, with a gradual ramp-up in the medium term. This means we do not plan to burden pilot participants with fees; initial deployments are funded by grants. As we move into months 6–18 (medium term), we introduce fees slowly – for instance, perhaps in the second year operators start paying a small subscription. This timing avoids deterring early adopters and allows the platform to prove its value before charging for it. By the end of 18 months, with multiple sites running, these fees should cover a meaningful portion of operating costs, and by long term (with dozens of sites), we aim for the platform to approach self-sufficiency on fee revenue alone for its core expenses.

However, we recognize that full financial sustainability may not be immediate. Thus, our model includes continued external support in parallel, especially to fund expansion and R&D. We anticipate **blended financing** to play a role – for example, using grant funding to subsidize operations in new regions until a critical mass of users is reached, or securing low-interest loans to cover infrastructure scaling which are paid back by later revenues. The Phase 2 report notes that post-pilot, financial contributions are expected from a combination of grants, in-kind

support, and future service contracts. In practice, this means we will keep pursuing grants (like challenge funds, climate innovation grants) to fund major upgrades or expansions, and we'll work with development partners who might chip in resources (e.g. a telco might provide SMS/USSD service at low cost, or a cloud provider might sponsor hosting for social good). Additionally, if large-scale impact investors come on board in later stages, their capital could help bridge any gaps as we grow the revenue base.

Our budgeting for sustainability also factors in prudent expense management. The use of Cardano blockchain technology, particularly Hydra, is chosen not just for technical fit but also for cost-efficiency (Hydra heads allow most transactions to be kept off the costly main chain, minimizing transaction fees). We also leverage open-source tools and existing infrastructure wherever possible to reduce costs. Over time, as the platform user base grows, economies of scale will improve our margins – hosting costs per site will drop, etc. We will continuously monitor the financial performance and adjust pricing or cost structure as needed to ensure we stay on a sustainable trajectory.

In conclusion, the financial sustainability model of Solar Village rests on a **hybrid approach**: developing steady operational revenue through subscriptions and usage fees (starting modestly and increasing with scale), tapping into new value streams like carbon credits, and securing institutional contracts for larger deployments – all while supplementing with strategic funding in the interim to fuel growth. This multi-pronged strategy will allow the project to maintain and expand its services without depending indefinitely on grants, ensuring that Solar Village can scale into a long-term, self-sustaining enterprise delivering social impact. The careful timing of monetization and the support of partners in the ecosystem give us confidence that the transition from a donor-backed pilot to a financially independent platform will be smooth and successful. Each new community added and each new revenue source unlocked brings us closer to the point where Solar Village's mission – lighting up villages with clean energy and transparent technology – is not just impactful, but also economically resilient.